Course Project

Code of Honor. All external resources used in the project, including research papers, open-source repositories, datasets, and any content or code generated using AI tools, e.g., ChatGPT, GitHub Copilot, Claude, Gemini, must be *clearly cited* in the final submission. The final report must also include *a clear breakdown of individual group member contributions.* Any lack of transparency in the use of external resources or in reporting group contributions will be considered academic dishonesty and will significantly impact the final evaluation.

Topic	Target-Speaker Speech-to-Text in Overlapping Speech
Category	Deep Learning for Audio Modelling

Objective Develop a target-speaker speech-to-text system that transcribes only a chosen speaker from audio containing overlapping speech. The system takes (i) a short enrollment utterance of the target speaker and (ii) a multi-speaker mixture, and outputs the target speaker's transcript. We will test this across 2-3 languages to see where language-specific modifications are needed.

Motivation Modern speech-to-text systems perform well on clean, single-speaker audio but degrade sharply when speakers overlap [4] [2]. This is a common occurrence in meetings, court proceedings, and media such as films or talk shows, and rather than attempting full diarization and separation, target-speaker transcription offers a more focused alternative: given a short enrollment clip, transcribe only the speaker of interest. Additionally, the rise of full-duplex speech-to-speech models is increasing the demand for labelled transcriptions of natural conversations with interruptions [1, 3]. A reliable target-speaker system could automatically generate such labels by isolating each participant and producing timed transcripts, even through overlapping speech.

Requirements The final submission should address the following requirements while implementation details are decided by the group members.

1. **Baseline ASR:** Implement ASR baselines from scratch or using pretrained open-source models and report WER on clean single-speaker audio and on overlapped mixtures.
2. **Overlapped-speech benchmark:** Build a data pipeline that generates controlled two-speaker (and optionally more) mixtures with configurable SNR and overlap ratio from open-source datasets.
3. **Target-speaker transcription:** Implement a method to transcribe a target speaker in mixtures given an enrollment clip, and demonstrate improved target-speaker WER relative to baseline.
4. **Robust evaluation:** Report WER versus SNR and overlap ratio for the different models. Include at least one measure of non-target leakage (e.g., rate of non-target words appearing in the output).

Milestones The following milestones are to be accomplished through the semester.

1. **Milestone 1:** Select datasets (English, optional: Twi and a third language), set up preprocessing and text normalization, and establish WER computation. Run pretrained ASR on clean speech as a baseline.

2. **Milestone 2:** Implement synthetic mixture generation (overlap + SNR control). Establish baseline results: ASR on mixtures without any separation/conditioning. Produce initial WER vs SNR/overlap plots.

3. **Milestone 3:** Implement target-speaker method: (a) compute speaker embeddings from enrollment audio using a pretrained speaker model, and (b) train a lightweight speaker-conditioned extraction or masking module, keeping the ASR model frozen.

4. **Milestone 4:** Perform full evaluation and ablations: WER versus enrollment audio length, non-target leakage analysis, and comparison across ASR model choices (pretrained, fine-tuned, trained from scratch). Optional: evaluate on Twi and additional languages to test cross-lingual generalization.

5. **Milestone 5:** Final Report and Presentation

Submission Guidelines The main body of work is submitted through Git. In addition, each group submits a final paper and gives a presentation. In this respect, please follow these steps.

- Each group must maintain a Git repository, e.g., GitHub or GitLab, for the project. By the time of final submission, the repository should have
  - Well-documented codebase
  - Clear `README.md` with setup and usage instructions
  - A `requirements.txt` file listing all required packages or an `environment.yaml` file with a reproducible environment setup
  - Demo script or notebook showing sample input-output
  - *If applicable,* a `/doc` folder with extended documentation

- A final report (maximum *5 pages*) must be submitted in a PDF format. The report should be written in the provided formal style, including an abstract, introduction, method, experiments, results, and conclusion.
  Important: Submissions that do not use template are considered *incomplete.*

- A 5-minute presentation (maximum *5 slides including the title slide*) is given on the internal seminar on Week 15, i.e., *Apr 6 to Apr 10,* by the group. For presentation, any template can be used.

Final Notes While planning for the milestones please consider the following points.

1. You are encouraged to explore innovative approaches to conditioning or generation as long as the core objectives are met.

2. While computational resources are limited, carefully chosen datasets and training setups can make even diffusion models feasible. Trade-offs, e.g., resolution, training steps, are expected and should be justified.

3. Teams are expected to manage their computing needs and are advised to perform early tests to estimate runtime and training feasibility. As graduate students, team members can use facilities provided by the university, e.g., ECE Facility. Teams are expected to inform themselves about the limitations of the available computing resources and design the model accordingly.

References

[1] Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour. Moshi: a speech-text foundation model for real-time dialogue, 2024.

[2] Naoyuki Kanda, Yashesh Gaur, Xiaofei Wang, Zhong Meng, and Takuya Yoshioka. Serialized Output Training for End-to-End Overlapped Speech Recognition. In Interspeech 2020, pages 2797–2801, 2020.

[3] Rajarshi Roy, Jonathan Raiman, Sang-gil Lee, Teodor-Dumitru Ene, Robert Kirby, Sungwon Kim, Jaehyeon Kim, and Bryan Catanzaro. Personaplex: Voice and role control for full duplex conversational speech models. 2026.

[4] Shinji Watanabe, Michael Mandel, Jon Barker, Emmanuel Vincent, Ashish Arora, Xuankai Chang, Sanjeev Khudanpur, Vimal Manohar, Daniel Povey, Desh Raj, David Snyder, Aswin Shanmugam Subramanian, Jan Trmal, Bar Ben Yair, Christoph Boeddeker, Zhaoheng Ni, Yusuke Fujita, Shota Horiguchi, Naoyuki Kanda, Takuya Yoshioka, and Neville Ryant. CHiME-6 Challenge: Tackling Multispeaker Speech Recognition for Unsegmented Recordings. In 6th International Workshop on Speech Processing in Everyday Environments (CHiME 2020), pages 1–7, 2020.